

Vol 10 Chapter 3 — ODEs and Sturm-Liouville Theory

Keeper (author pass — deep math/physics revision)

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Chapter 3 — ODEs and Sturm-Liouville Theory

Level 1 — one sentence

Ordinary differential equations describe physical systems with one independent variable, and Sturm-Liouville theory gives the spectral framework for boundary-value problems whose eigenfunctions form complete orthogonal bases — the foundation for separation-of-variables solutions to many physics PDEs (Schrödinger, heat, wave).

Level 2 — graduate-physicist precision

3.1 First-order ODEs

$y' = f(x, y)$. Picard-Lindelöf: if f is Lipschitz in y , unique solution exists locally.

Linear: $y' + p(x)y = q(x)$ — integrating factor solution.

Separable: $y' = g(x)h(y)$ — $\int dy/h = \int g(x)dx$.

3.2 Second-order linear ODEs

$y'' + p(x)y' + q(x)y = r(x)$. General solution = particular + homogeneous.

Constant-coefficient: $y'' + ay' + by = 0$. Characteristic equation $\lambda^2 + a\lambda + b = 0$; solutions $e^{\lambda_1 x}, e^{\lambda_2 x}$ (or $e^{\lambda x}, xe^{\lambda x}$ for repeated root).

Frobenius method for regular singular points.

3.3 Sturm-Liouville problems

Self-adjoint form: $[p(x)y']' + [q(x) + \lambda w(x)]y = 0$ on interval $[a, b]$ with boundary conditions.

Sturm-Liouville theorem: - Eigenvalues λ_n are real, discrete, bounded below - Eigenfunctions ϕ_n are orthogonal with weight w : $\int_a^b \phi_m \phi_n w dx = 0$ for $m \neq n$ - Eigenfunctions form complete basis (any function expandable as $f = \sum c_n \phi_n$)

Examples: - Fourier series: $-y'' = \lambda y$, periodic BC $\rightarrow \sin(n\pi x/L), \cos(n\pi x/L)$ - Legendre: $((1 - x^2)y')' = -\lambda y$ on $[-1, 1] \rightarrow$ Legendre polynomials $P_\ell(x)$, $\lambda = \ell(\ell + 1)$ - Bessel: gives Bessel functions - Hermite: gives Hermite polynomials (HO eigenfunctions, Vol 5 Ch 2)

3.4 Connection to substrate spectral theory

The substrate Hamiltonian (Vol 5 Ch 4) is the Casimir on $L^2(D_{IV}^5; L_\lambda)$ — a Sturm-Liouville-type spectral problem in higher dimensions. K-types are the eigenfunctions; Casimir eigenvalues are the λ_n ; substrate K-type orthogonality is the higher-dimensional Sturm-Liouville orthogonality.

3.5 K-audit anchors

- Vol 5 Ch 4: substrate Casimir as Sturm-Liouville analog

Level 3 — 5th-grader accessibility

ODEs: equations involving derivatives of an unknown function. Linear constant-coefficient: solve by guessing $e^{\lambda x}$. Sturm-Liouville problems: special class of eigenvalue ODEs whose eigenvalues are discrete real numbers and whose eigenfunctions form a complete orthogonal set. Examples: Fourier, Legendre, Bessel, Hermite functions. These are the building blocks for separation-of-variables solutions to physics PDEs.

What comes next

Chapter 4 develops PDEs.

Where to look this up

- Courant and Hilbert *Methods of Mathematical Physics*
- BST: Vol 5 Ch 4